KAIST
2026 Special Topics in Physics Quantum Electronic Devices I: Interferometers and Fractional Particles
Homework 1

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#1

Assume $n(t)$ obeys the diffusion equation and the initial condition $n(x,0) = \delta(x)$, Then we can solve the diffusion equation by Fourier transform as follows:

$$\partial_t n(x,t) = D\nabla^2 n(x,t) \implies \partial_t \tilde{n}(k,t) = -Dk^2 \tilde{n}(k,t) \tag{1}$$

$$\tilde{n}(k,t) = \tilde{n}(k,0)e^{-Dk^2 t} = e^{-Dk^2 t} \tag{2}$$

$$n(x,t) = \frac{1}{(2\pi)^d}\int e^{ik\cdot x}e^{-Dk^2 t}dk = \frac{1}{(4\pi Dt)^{d/2}}\exp\left(-\frac{x^2}{4Dt}\right) \tag{3}$$

Due to the conservation law of diffusion equation, the denominator is 1:

$$\int n(x,t)dx = 1 \tag{4}$$

And the mean absolute displacement can be evaluated as follows:

$$\langle |x| \rangle = \int |x| n(x,t)dx = \frac{1}{(4\pi Dt)^{d/2}}\int_0^\infty r\exp\left(-\frac{r^2}{4Dt}\right)\Omega_d r^{d-1}dr = \frac{2\Gamma(\frac{d+1}{2})}{\Gamma(\frac{d}{2})}\sqrt{Dt} \tag{5}$$

Where Ω_d is the surface area of the unit sphere in d-dimension.

Hence,

$$l(t) \propto \sqrt{Dt} \tag{6}$$

#2

The total number of particle can be evaluated by integrating the density of states as follows:

$$N = \sum_{\frac{2\pi}{L}\sqrt{n_x^2+n_y^2}\le k_F} \sum_{\sigma} 1 \approx \int_{k_x^2+k_y^2<=k_F^2} g_s \left(\frac{L}{2\pi}\right)^2 dk_x dk_y = \frac{L^2 g_s k_F^2}{4\pi} \tag{7}$$

Using the relation $L \times L = V$ and $n_e = N/V$,

$$n_e = \frac{g_s k_F^2}{4\pi} \tag{8}$$

#3

Assume that the dispersion relation of the system is given by $E(k) = f(|k|)$. Then the density of states per area can be derived as follows:

$$\sum_{\sigma} \left(\frac{L}{2\pi}\right)^2 dk_x dk_y = 2g_s \pi k \left(\frac{L}{2\pi}\right)^2 dk = \frac{2g_s \pi k}{f'(k)} \left(\frac{L}{2\pi}\right)^2 dE = A\rho(E)dE \tag{9}$$

$$\implies \rho(E) = \frac{g_s k}{2\pi f'(k)} \tag{10}$$

For the quadratic dispersion relation, $f(k) = \frac{\hbar^2 k^2}{2m}$,

$$\rho(E) = \frac{g_s k}{2\pi f'(k)} = \frac{g_s m}{2\pi \hbar^2} \tag{11}$$

For the linear dispersion relation, $f(k) = \hbar v k$,

$$\rho(E) = \frac{g_s k}{2\pi f'(k)} = \frac{g_s k}{2\pi \hbar v} = \frac{g_s E}{2\pi (\hbar v)^2} \tag{12}$$